Supplement — Aggregation-aware monthly CHD/HF thermal associations, Hong Kong, 2013–2023

Companion to chd_hf_thermal_associations_2013_2023.md. All health-effect numbers below are drawn from outputs/release_chd_hf/ unless labelled SYNTHETIC_CALIBRATION or REAL_PUBLIC_HKO. Provenance for CHD/HF association rows is HA_APPROVED_AGGREGATE.

S0. Reporting checklist

Item	Location
Estimand (monthly count ratio; days offset)	Main Methods; S2
Outcome construction (first recorded hospitalisation after diagnosis; column-label semantics open)	Main Methods; S1
Complete 12-contrast continuity table	outputs/release_chd_hf/tables/table2_core_models.csv
Claim ledger identity CVD-01 to CVD-12	outputs/release_chd_hf/tables/claim_ledger_v2.csv
SE ladder Model/HC1/NW3/NW6	outputs/release_chd_hf/tables/table4_uncertainty_ladd Figure 5
Multiplicity (BH q across 12)	Table 2 q_value_core_bh
Residual ACF / Ljung–Box	S6; out- puts/release_chd_hf/supplement/chd_pathway_core_diagn out- puts/release_chd_hf/supplement/hf_pathway_core_diagno
Joint vs separate / VIF	S3
Trend, depletion, lag, influence, unoffset INGARCH	S5
Weather source lock + five reproduction checks	S4
M|D calibration F1.0 / F1.1 / F1.2	S7
Provenance firewall	S8
Reproducibility commands	S9
Scope exclusions	S10
Authorship / Gate 3 / ICD / stroke open items	authorship_and_governance_decisions.md

S1. Outcome and exposure definitions

S1.1 Cohort and event

- Cohort universe: people diagnosed with type 2 diabetes and/or hypertension during 2013–2023.
- Event: first recorded hospitalisation after the patient's first diagnosis record for CHD or HF.
- Admission cause: not recorded.

- Grain: territory × calendar month, January 2013–December 2023 (132 months).
- Observed columns (gitignored source): CHD `chd_inpatient`; HF `hf_inpatient`.
- Care-setting label: inpatient semantics are inferred from column names; confirmation remains human-owned.
- Stroke: named in covering correspondence; file not attached; no stroke estimates.

S1.2 Outcome totals

From `outputs/release_chd_hf/tables/table1_outcome_summary.csv`:

Outcome	Months	Total events	Monthly mean
CHD	132	156,156	1,183.0
HF	132	29,681	224.9

S1.3 Thermal exposures used in the continuity panel

Internal pathway IDs are retained here for archive mapping only; they do not appear in the journal scientific body.

Pathway ID	Exposure	Contrast
P01A	Monthly mean temperature	per 1 °C
P02A	Monthly mean of daily maximum temperature	per 1 °C
P02B	Monthly mean of daily minimum temperature	per 1 °C
P04A	Official hot nights (`Tmin ≥ 28 °C`)	per 5 days
P04B	Official cold days (`Tmin ≤ 12 °C`)	per 5 days
P04C	Official very hot days (`Tmax ≥ 33 °C`)	per 5 days

S1.4 Denominator

Analysis-of-record offset: `log(days in month)`. Population aged 35+ × days is an ecological sensitivity only. It is not person-time among members of the diabetes or hypertension cohort still at risk of a first CHD or HF hospitalisation.

S2. Complete model table index

Table / figure	Path	Role
Table 1	`outputs/release_chd_hf/tables/table1_outcome_summary.csv`	Outcome summary
Table 2	`outputs/release_chd_hf/tables/table2_continuity_NW6.csv`	Twelve continuity model contrasts
Table 3	`outputs/release_chd_hf/tables/table3_robustness_summary.csv`	Robustness ranges
Table 4	`outputs/release_chd_hf/tables/table4_SE_uncertainty_ladder.csv`	12 × 4 SE ladder
Claim ledger v2	`outputs/release_chd_hf/tables/claim_ledger_v2.csv`	CVD-01 to CVD-12 with tiers
Analysis availability	`outputs/release_chd_hf/tables/analysis_availability.csv`	Scope / blocker inventory
Figure 1	`outputs/release_chd_hf/figures/figure1_indexed_outcome_series.png`	Indexed series
Figure 2	`outputs/release_chd_hf/figures/figure2_seasonal_pattern.png`	Seasonality
Figure 3	`outputs/release_chd_hf/figures/figure3_core_forest.png`	Continuity forest
Figure 4	`outputs/release_chd_hf/figures/figure4_trend_depletion_sensitivity.`	Trend/depletion
Figure 5	`outputs/release_chd_hf/figures/figure5_se_method_ladder.png`	SE ladder forest

S2.1 Pathway panel archives

- `outputs/release_chd_hf/supplement/combined_pathway_panel_estimates.csv`
- `outputs/release_chd_hf/supplement/chd_pathway_panel_forest.png`
- `outputs/release_chd_hf/supplement/hf_pathway_panel_forest.png`
- Core diagnostics: `outputs/release_chd_hf/supplement/chd_pathway_core_diagnostics.csv`, `outputs/release_chd_hf/supplement/hf_pathway_core_diagnostics.csv`
- Core fit summary: `outputs/release_chd_hf/supplement/cvd_core_model_fit.csv`
- Robust SE/offset/family ladder: `outputs/release_chd_hf/supplement/cvd_core_robust_estimates.csv`

Joint extreme-day and nested heat-month structures remain exploratory. Separate continuity IDs P01A, P02A/B, and P04A–C map to the manuscript continuity panel.

Legacy pollution, humidity, and influenza pathway fits in the broader archive are not adjusted versions of these separate continuity models.

S3. Joint versus separate exposures and collinearity

File: `outputs/release_chd_hf/supplement/cvd_single_vs_joint_estimates.csv` VIF: `outputs/release_chd_hf/supplement/cvd_exposure_vif.csv` Exposure correlations: `outputs/release_chd_hf/supplement/cvd_exposure_correlations.csv`; Figure S1 `outputs/release_chd_hf/supplement/figureS1_exposure_correlation.png`

After month and trend residualisation:

- VIF for mean maximum and minimum temperature (joint temperature model) = 4.66.
- VIF for hot nights and very hot days (joint extreme-day model) ≈ 1.96.
- VIF for cold days in the joint extreme-day set ≈ 1.00.

Under days-only offset and Newey–West lag-6 intervals, the CHD hot-night count ratio was 1.022 in the separate hot-night model and 1.045 (1.015–1.075) in the joint extreme-day model. The HF cold-day coefficient was about 1.073 in both structures. Joint inflation of the CHD hot-night coefficient is retained as a collinearity warning.

S4. Source-validation audit (weather)

Provenance: `REAL_PUBLIC_HKO`. Files:

- `outputs/release_chd_hf/supplement/methods_feasibility/weather_definition_source_lock.csv`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_definition_source_validation.`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_event_morphology_audit.csv`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_spillover_exposure_build.md`
- Spillover figure: `outputs/release_chd_hf/supplement/methods_feasibility/figure_spillover_event`

S4.1 Five source reproduction checks (all pass)

validation_id	expected	observed	source DOI
`li_events_1980_2023`	57	57	10.1016/j.atmosres.2024.107845
`wang_vhd_days_2006_2015`	204	204	10.1016/j.scitotenv.2019.07.039
`wang_hn_days_2006_2015`	230	230	10.1016/j.scitotenv.2019.07.039
`wang_vhd_ge5_event_days_2006_2015`	48	48	10.1016/j.scitotenv.2019.07.039
`wang_hn_ge5_event_days_2006_2015`	56	56	10.1016/j.scitotenv.2019.07.039

Release validation recorded `weather_source_validation_all_pass = TRUE` (`outputs/reports/cvd_real_rele`

S4.2 Provisional hot-month / cold-month calendar

Provisional HM/CM flags remain unlocked. Calendar figure: `outputs/release_chd_hf/supplement/figureS2` Estimates in `outputs/release_chd_hf/supplement/chd_hm_cm_panel_estimates.csv` and `outputs/release_chd_hf/supplement/hf_hm_cm_panel_estimates.csv` are not eligible for primary claims until weather reference rules are locked by the weather co-investigator.

S5. Sensitivity ladders

S5.1 Offset and family

File: `outputs/release_chd_hf/supplement/cvd_core_robust_estimates.csv`. For each continuity-panel exposure: offsets (days only, population × days, none); families (negative binomial, quasi-Poisson); SE methods (model, HC1, NW3, NW6). Manuscript Table 2 freezes the days-only / negative-binomial / NW6 continuity slice; Table 4 shows the SE ladder without selecting an interval by null exclusion.

S5.2 Trend and first-event depletion

File: `outputs/release_chd_hf/supplement/cvd_trend_depletion_sensitivity.csv`; summary ranges in `outputs/release_chd_hf/tables/table3_robustness_summary.csv`; Figure 4.

CHD hot nights: baseline 1.022; scenario range approximately 1.011–1.025. HF cold days: baseline 1.073; scenario range approximately 1.043–1.113; pre-2020 1.113 (1.053–1.176).

S5.3 Lag

File: `outputs/release_chd_hf/supplement/cvd_lag_sensitivity.csv`.

Outcome	Exposure	Lag 0	Lag 1	Lag 2
CHD	Hot nights /5	1.022 (1.002–1.042)	1.010 (0.979–1.042)	1.016 (0.987–1.045)
HF	Cold days /5	1.073 (1.006–1.144)	1.073 (1.014–1.135)	1.053 (0.985–1.127)

Lag-one-month temperature is a labelled sensitivity. It is not a recovered daily lag curve.

S5.4 Influence

File: `outputs/release_chd_hf/supplement/cvd_influence_sensitivity.csv`.

Outcome	Pathway ID	Excluded month	RR after exclusion	Max Cook's D
CHD	P04A	2020-02	1.021 (1.001–1.040)	0.110
HF	P04B	2022-02	1.088 (1.034–1.144)	0.085

S5.5 Count-series dependence (unoffset INGARCH)

File: `outputs/release_chd_hf/supplement/cvd_count_timeseries_sensitivity.csv`.

Negative-binomial INGARCH(1,1) models were fitted **without an offset**. They reduced residual ACF1 (CHD hot-night residual ACF1 ≈ 0.25; HF cold-day residual ACF1 ≈ −0.04) while retaining positive CHD hot-night and HF cold-day associations. These fits are unoffset residual-dependence diagnostics. They are not directly comparable with the days-offset continuity estimates and are not a second primary estimand.

S6. Residual diagnostics

Outcome	Continuity residual ACF1 (approx.)	Range across six continuity exposures	Ljung–Box lag 6
CHD	0.51	0.508–0.534	Rejects white noise
HF	0.15	0.131–0.179	Does not reject

Plots: `outputs/release_chd_hf/supplement/chd_core_residual_acf_pacf.png`, `outputs/release_chd_hf/supplement/hf_core_residual_acf_pacf.png`. Tables: `outputs/release_chd_hf/supp`
`outputs/release_chd_hf/supplement/hf_pathway_residual_acf.csv`.

S7. M|D calibration scenarios and gates

Provenance: `SYNTHETIC_CALIBRATION` only. Quarantine folder: `outputs/release_chd_hf/supplement/metho`
These materials validate methods. They are not CHD/HF health findings.

S7.1 Rationale

Monthly sums erase within-month timing. The Basagaña–Ballester aggregated likelihood provides a constrained route from daily exposure to monthly counts (Basagaña and Ballester 2024, 2026). Admission of real daily coefficients requires synthetic calibration gates to pass under the frozen protocol (`analysis_plan/final_reanalysis_protocol_2026-08-10.md`, Section 8).

S7.2 F1.0 pilot failure

Artifact: `outputs/release_chd_hf/supplement/methods_feasibility/md_pilot_f1_0_failure_note.md`.
Design: 100 replicates; daily nuisance seasonality = one annual sine/cosine pair. CHD-like null cells were acceptable (Type I ≈ 0.00–0.06; coverage ≈ 0.94–1.00). HF-like null cells failed under every kernel (Type I ≈ 0.62–0.75; coverage ≈ 0.25–0.38). Interpretation: model-design failure from inadequate nuisance seasonality, not evidence of a real HF cold effect.

S7.3 F1.1 correction

Amendment: replace the harmonic with calendar-month indicators, matching the analysis-of-record seasonality, while remaining inside the 20-parameter cap. The failed F1.0 pilot remains auditable. No new real health model was fitted to choose the correction.

S7.4 F1.2 worst-cell decision (500 replicates)

Artifact: `outputs/release_chd_hf/supplement/methods_feasibility/md_calibration_f1_2_decision_repo`
Decision: `FAIL_METHODS_FEASIBILITY_ONLY`. All gates passed: FALSE.

Gate	Passed	Value	Requirement
Convergence min	Yes	1.000	≥0.95
Null Type I range	No	0.048–0.150	within 0.03–0.08
Coverage min	No	0.840	≥0.90
Relative bias max	No	32.77	≤0.20
False-sign max	No	0.808	≤0.10
Hessian condition	Yes	1.82×10^3	$<1\times10^8$
Divergent fits	Yes	0	<0.05
Stress coverage min	No	0.842	≥0.85

Method-validation figures (not health findings):

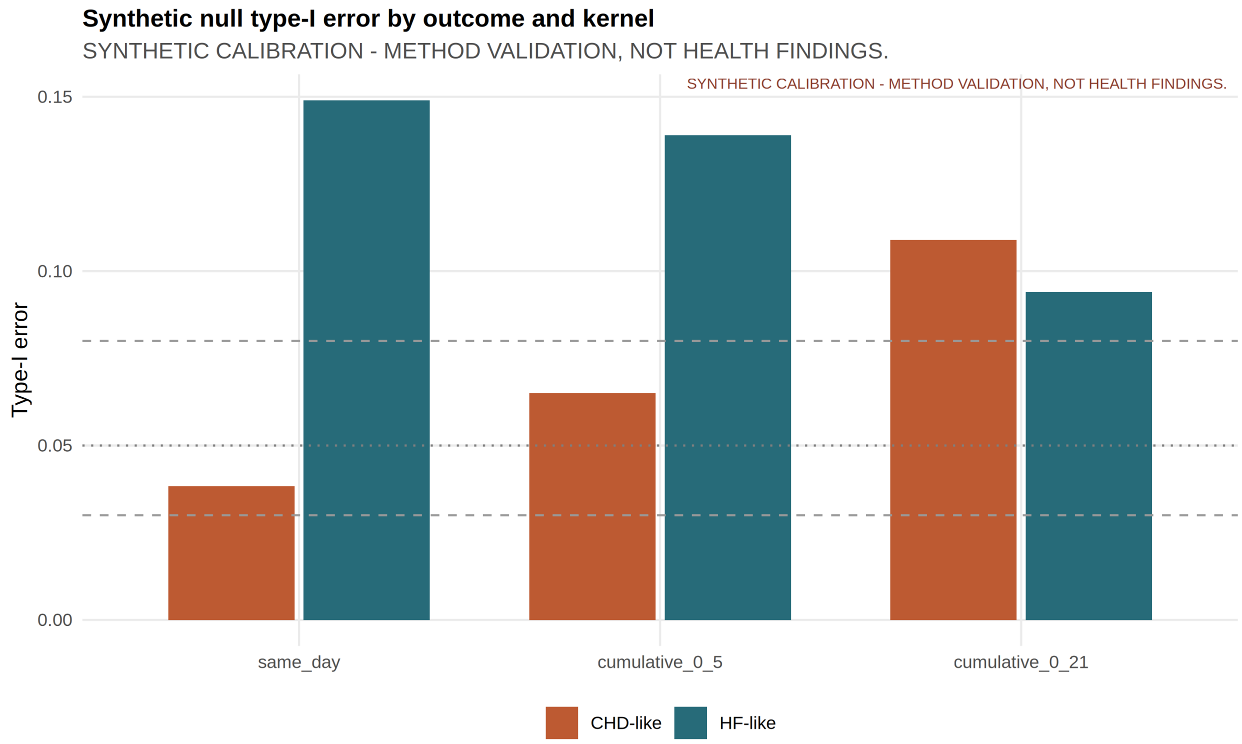

Figure 1: Methods validation only — synthetic M|D null Type I by outcome and kernel. Not a CHD/HF health result.

Paths:

- `outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_type1_by_outcome_k`
- `outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_coverage_by_outcom`
- `outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_gate_passfail.png`

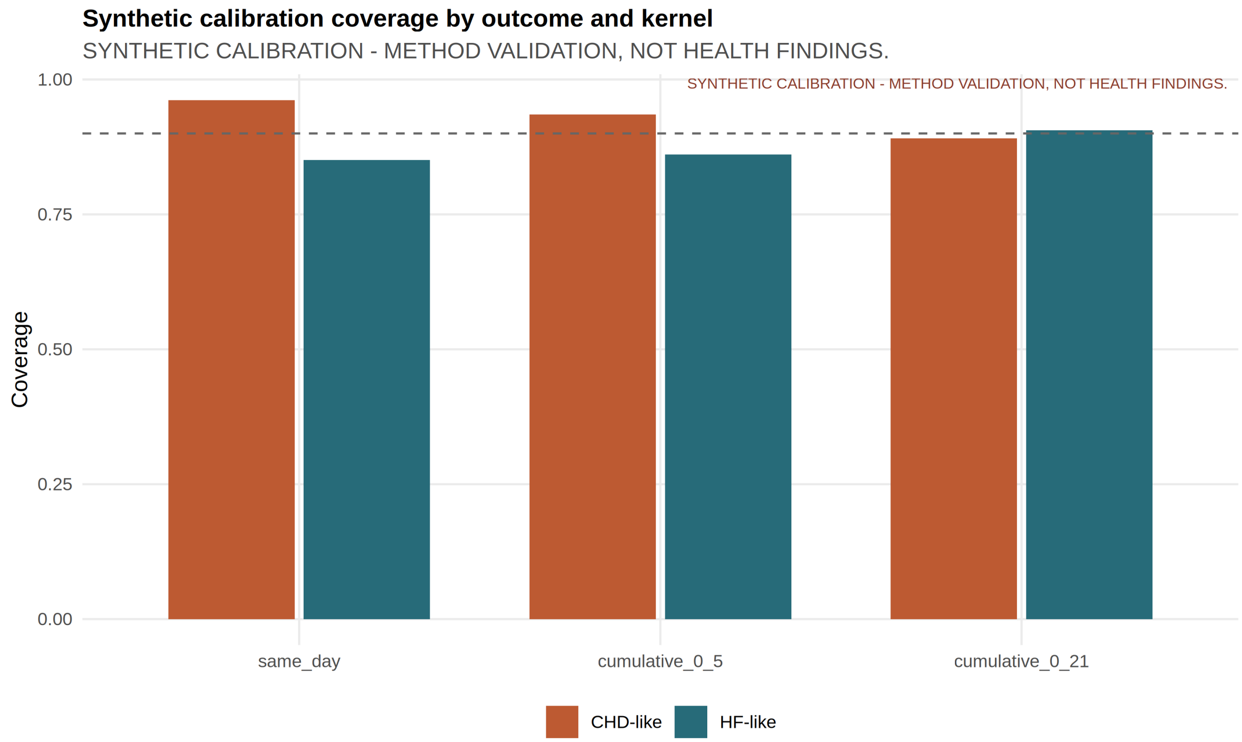

Figure 2: Methods validation only — synthetic M|D coverage by outcome and kernel. Not a CHD/HF health result.

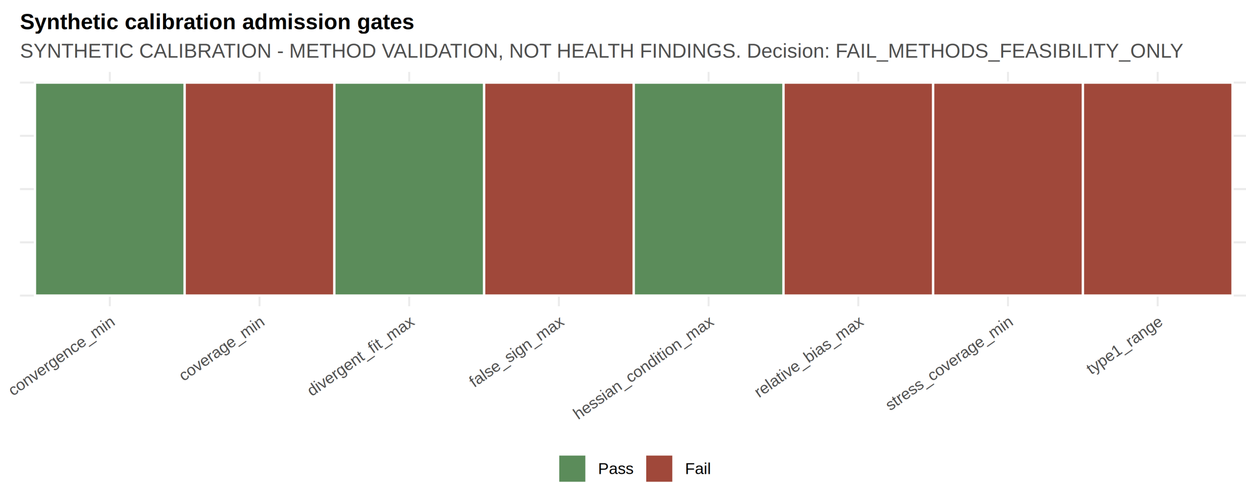

Figure 3: Methods validation only — F1.2 gate pass/fail summary. Not a CHD/HF health result.

S7.5 Real M|D admission rule

Because F1.2 calibration failed, no real daily-exposure coefficient is eligible for manuscript Results. Real constrained M|D fitting additionally requires governed analysis panels that are not redistributed in the public repository.

S8. Provenance firewall

Class	Label	May appear on main claim surfaces?
Governed CHD/HF aggregates and model summaries	`HA_APPROVED_AGGREGATE`	Yes
Public HKO weather validation	`REAL_PUBLIC_HKO`	Yes, as exposure methods
Synthetic calibration	`SYNTHETIC_CALIBRATION`	Supplement methods-feasibility only
Daily mortality AF (Jingwen Liu 2020)	Literature	Introduction/Discussion only; not morbidity coefficients
Modelled excess deaths (Zhenyuan Liu 2026 medRxiv)	Literature / preprint	Companion baseline only
Source HA xlsx / merged panels	Governed; gitignored	Never committed
Private correspondence PDF	Internal provenance	Not cited; no raw addresses or private quotations

Forbidden promotions (from `analysis_plan/final_claim_tiers.yml`):

- calibration → real finding;
- exploratory → primary without human headline freeze;
- mortality AF → monthly morbidity effect;
- modelled excess death → observed hospitalisation burden;
- general-population offset → cohort incidence rate.

S9. Reproducibility

Identity checks already recorded:

1. Every Table 2 row matches `outputs/release_chd_hf/tables/claim_ledger_v2.csv` for RR, CI, p, q, SE method, and provenance.
2. BH q-values recomputed with `p.adjust(..., method="BH")` across the twelve NW6 p-values.
3. Release validation: `outputs/reports/cvd_real_release_validation.md` — 28/28 checks passed on 10 August 2026.
4. Manifest hashes: `outputs/release_chd_hf/release_manifest.csv`.
5. Index: `outputs/release_chd_hf/RELEASE_INDEX.md`.

S9.1 Commands for completed public / release / calibration checks

From the repository root:

```
# Public HKO spillover morphology + five source reproduction checks
Rscript scripts/35_build_spillover_exposures.R
```

```
# Rebuild disclosure-minimised release package from existing analysis tables
Rscript scripts/38_cns_final_release_artifacts.R

# Recompute F1.2 worst-cell gates from existing synthetic raw metrics
# (does not refit real health models; requires outputs/calibration_md/md_calibration_raw_metric
Rscript scripts/45_recompute_md_calibration_gates.R

# Release identity / provenance checks (tables, claim ledger, SE ladder, quarantine policy)
Rscript scripts/34_cvd_real_release_checks.R
```

S9.2 Governed full rerun (local files required)

A complete real-data pathway rerun requires governed monthly CHD/HF aggregates and merged analysis panels that are gitignored and not present in the public checkout:

```
# Requires local governed inputs under data_raw/ha_secure_placeholder/ (or equivalent)
# and a constructed analysis panel. Do not commit HA microdata.
PATHWAY_MODE=real OUTCOMES=chd,hf Rscript scripts/run_cvd_full_analysis.R
```

Without those local governed files, treat release tables under `outputs/release_chd_hf/` as the analysis-of-record artifact set for manuscript drafting.

S10. Scope exclusions

Excluded from the journal scientific claims by design or delivery:

- Stroke results (file not delivered);
- Age/sex or medication effect modification (strata not delivered);
- Principal-diagnosis CHD/HF or AMI claims (admission cause absent);
- Locked hot-month / cold-month primary estimates (weather reference rules unlocked);
- Cohort incidence with the true still-at-risk denominator (denominator unavailable);
- Real daily-exposure M|D coefficients (calibration failed);
- Confounding-adjusted continuity estimates for pollution, humidity, or influenza (legacy/joint archive paths only).

Cloud-runtime availability of governed panels is an internal operations fact and is documented in the supervisor integrated report, not as a journal result.

S11. What this supplement does not contain

- Source monthly HA count files
- Merged health–environment panels with event counts
- Synthetic coefficients presented as health findings
- Private email text or raw addresses
- Final author order or Gate 3 freeze

References

Basagaña, Xavier, and Joan Ballester. 2024. "Unbiased Temperature-Related Mortality Estimates Using Weekly and Monthly Health Data: A New Method for Environmental Epidemiology and Climate Impact Studies." The Lancet Planetary Health 8 (10): e766–77. https://doi.org/10.1016/S2542-5196(24)00212-2.

———. 2026. “Unbiased Estimates Using Temporally Aggregated Outcome Data in Time Series Analysis: Generalization to Different Outcomes, Exposures, and Types of Aggregation.” Epidemiology 37 (1): 16–20. https://doi.org/10.1097/EDE.0000000000001923.